## A short kernel vector cannot bound an arbitrary index

$$\Lambda N = \{(x,y) \in \mathbb{Z}^2 : x - y \equiv 0 \bmod N\}$$

| $N = 3$ | $N = 30$ | $N = 3{,}000$ |
|---|---|---|
| index = 3 | index = 30 | index = 3,000 |
| $(1,1) \in \Lambda N$ | $(1,1) \in \Lambda N$ | $(1,1) \in \Lambda N$ |

The actual logarithmic rows need arithmetic rigidity
A generic lattice estimate does not supply it